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Student Number

Mathematics Extension 1

2020 YEAR 11 ASSESSMENT TASK 2

Staff Involved:

DZP* BHC

ARP PDJ

PJR ARM

FRIDAY 26TH JUNE

PERIOD 2 OR 5

110 copies

General Instructions

- Working time – 50 minutes
- Write using blue or black pen
- Write your student number in the space provided at the top of this page
- NESA approved calculators may be used
- A reference sheet is provided
- Show ALL your working
- Marks may not be awarded for careless or badly arranged working
- Answer ALL questions in the spaces provided
- Marks are as indicated on the paper

Total Marks: 56

Topics covered:

- Functions and graphs
- Trigonometry
- Logarithms and exponentials
- Further graphs
- Differentiation

Marks

Question 1 (4 marks)

Given that $\log_a 3 = 1.7$ and $\log_a 5 = 2.5$, find the value of:

(a) $\log_a(15a)$

2

(b) $\log_a\left(\frac{9}{5}\right)$

2

Question 2 (5 marks)

Given that $f(x) = 3x^2 - x$:

(i) find $f(x + h)$ in expanded simplified form.

2

(ii) use $\lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right)$ to find $f'(x)$ from first principles.

3

Question 3 (5 marks)

(i) Show that $\frac{\sin^2 \theta}{1-\cos \theta} - \frac{\sin^2 \theta}{1+\cos \theta} = 2 \cos \theta$ **3**

(ii) Hence, solve $\frac{\sin^2 \theta}{1-\cos \theta} - \frac{\sin^2 \theta}{1+\cos \theta} = 1$, $-90^\circ \leq \theta \leq 90^\circ$ **2**

Question 4 (6 marks)

Differentiate the following:

(a) $y = 1 - 3x^2$ **1**

(b) $f(x) = \frac{4}{\sqrt{x}} - \sqrt[3]{x}$ **2**

(c) $y = x^2\sqrt{1-x}$ **3**

Question 5 (8 marks)

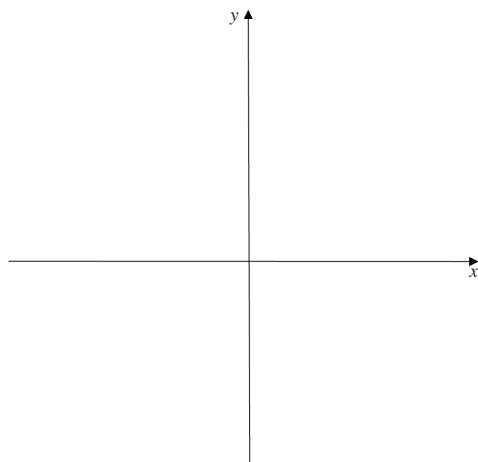
Consider the function $f(x) = \frac{2x}{x^2-1}$.

(i) Show, with necessary working, whether it is an odd function, even function or neither. **2**

(ii) Find the equation(s) of any vertical asymptotes. **1**

(iii) By first finding $\frac{d}{dx} \frac{2x}{x^2-1}$ show that the derivative is always negative. **3**

(iv) Sketch the function, showing all asymptotes and intercepts. **2**



Question 6 (6 marks)

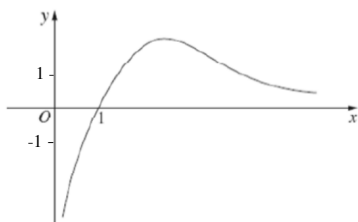
Solve for x :

(a) $\log_{10}(x+4) - \log_{10}(x-5) = 1$ **3**

(b) $2(4^x) - 17(2^x) + 8 = 0$ **3**

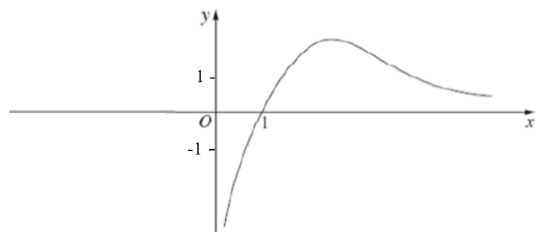
Question 7 (6 marks)

The graph below shows the function of $y = g(x)$.



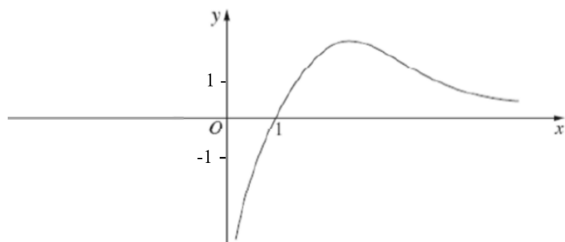
- (i) Sketch the graph of $|y| = g(x)$.

2



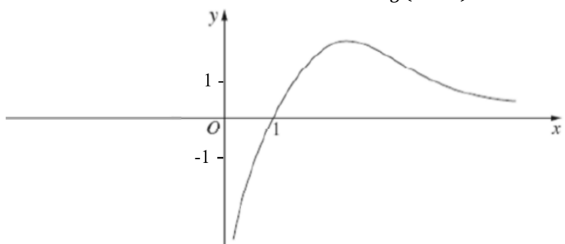
- (ii) Sketch the graph of $y = \frac{1}{g(x)}$.

2



- (iii) Sketch the graph of $y = f(x)$, where $f(x) = \begin{cases} g(x) & x \geq 1 \\ g(2-x) & x < 1 \end{cases}$

2



Question 8 (7 Marks)

- (a) Find the equation of the **normal** to the curve $y = \frac{1}{1+x^2}$ at the point on the curve where $x = 1$.

4

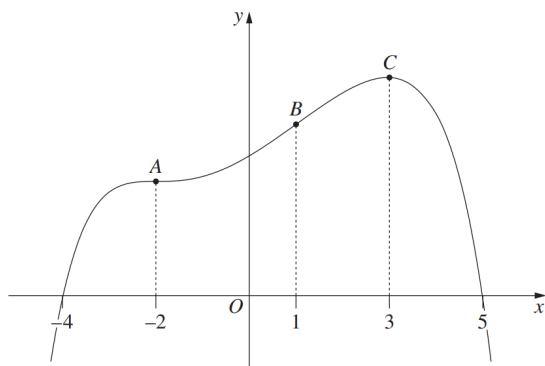
- (b) At what points on the curve $y = x^3 - 6x^2 + 2x - 9$ are the tangents parallel to the line $7x + y = 4$?

3

Question 9 (3 marks)

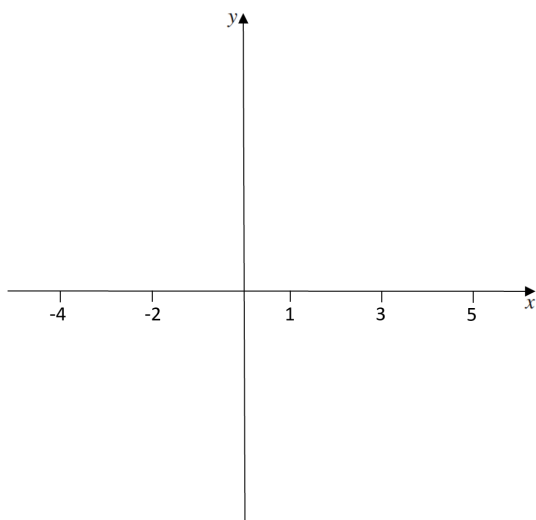
The diagram shows the graph of a function $f(x)$.

The graph has horizontal gradients at points A and C. Between points A and C, the gradient is steepest at point B.



Sketch the graph of the derivative $f'(x)$ on the axis below.

3

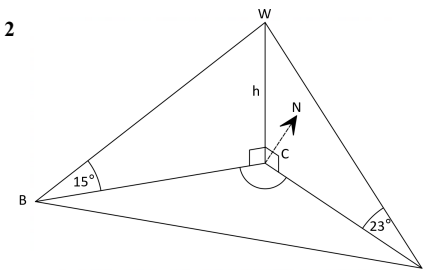


Question 10 (6 marks)

A man walking along a straight road on a flat floodplain stops at point A and notices a windmill on a bearing of 300° . He measured the angle of elevation to the top of the windmill (W) to be 23° . After walking a further 200m along the straight road, he stops at point B and notices the windmill is now on a bearing of 025° . He measured the angle of elevation to the top of the windmill to be 15° .

(i) Show that $\angle ACB$ is 85° .

2



(ii) Show the length of BC is $htan75^\circ$.

1

(iii) Calculate the height, h , of the windmill to 1 decimal place.

3

Q1, Year 11 Assessment Task 2 Friday 26th June

$$\begin{aligned}
 \text{(a) } \log_a(15a) &= \log_a 15 + \log_a a \\
 &= \log_a 3 + \log_a 5 + \log_a a \\
 &= 1.7 + 2.5 + 1 \\
 &= 5.2
 \end{aligned}$$

$$\begin{aligned}
 \text{(b) } \log_a\left(\frac{9}{5}\right) &= \log_a 3^2 - \log_a 5 \\
 &= 2(\log_a 3) - \log_a 5 \\
 &= 2 \times 1.7 - 2.5 \\
 &= 0.9
 \end{aligned}$$

$$\begin{aligned}
 \text{Q2, (i) } f(x+h) &= 3(x+h)^2 + (x+h) \\
 &= 3x^2 + 6xh + 3h^2 + x + h
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } f'(x) &= \lim_{h \rightarrow 0} \left[\frac{3x^2 + 6xh + 3h^2 + x + h - (3x^2 + x)}{h} \right] \\
 &= \lim_{h \rightarrow 0} \left[\frac{6xh + 3h^2 + h}{h} \right] \\
 &= \lim_{h \rightarrow 0} [6x + 3h + 1] \\
 &= 6x + 1
 \end{aligned}$$

Q3, (i)

$$\text{LHS} = \frac{\sin^2 \theta}{1 - \cos \theta} - \frac{\sin^2 \theta}{1 + \cos \theta}$$

$$= \frac{\sin^2 \theta (1 + \cos \theta) - \sin^2 \theta (1 - \cos \theta)}{(1 - \cos \theta)(1 + \cos \theta)}$$

$$= \frac{2 \sin^2 \theta \cos \theta}{1 - \cos^2 \theta}$$

$$= \frac{2 \sin^2 \theta \cos \theta}{\sin^2 \theta}$$

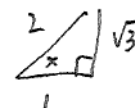
$$= 2 \cos \theta$$

$$= \text{RHS}$$

$$\text{(ii) Solve } \frac{\sin^2 \theta}{1 - \cos \theta} - \frac{\sin^2 \theta}{1 + \cos \theta} = 1$$

$$\therefore 2 \cos \theta = 1$$

$$\cos \theta = \frac{1}{2}$$



$$\theta = -60^\circ, 60^\circ$$

Q4,

$$(a) y = 1 - 3x^2$$

$$\frac{dy}{dx} = -6x$$

$$(b) y = \frac{4}{\sqrt{x}} - \sqrt[3]{x}$$

$$\frac{dy}{dx} = 2x^{-\frac{1}{2}} - \frac{1}{3}x^{-\frac{2}{3}}$$

$$(c) y = x^2 \sqrt{1-x}$$

$$\frac{dy}{dx} = x^2 \cdot \frac{1}{2}(1-x)^{-\frac{1}{2}} + \sqrt{1-x} \times 2x$$

$$\frac{dy}{dx} = 2x\sqrt{1-x} - \frac{x^2}{2\sqrt{1-x}}$$

$$Q5, f(x) = \frac{2x}{x^2-1}$$

$$\begin{aligned} (i) f(-x) &= \frac{2(-x)}{(-x)^2-1} \\ &= \frac{-2x}{x^2-1} \\ &= -\left[\frac{2x}{x^2-1}\right] \\ &= -f(x) \end{aligned}$$

$\therefore$ an odd function

Q5, (ii) Vertical asymptotes at $x=1$ and $x=-1$.

$$(iii) \frac{d}{dx} \left[\frac{2x}{x^2-1} \right]$$

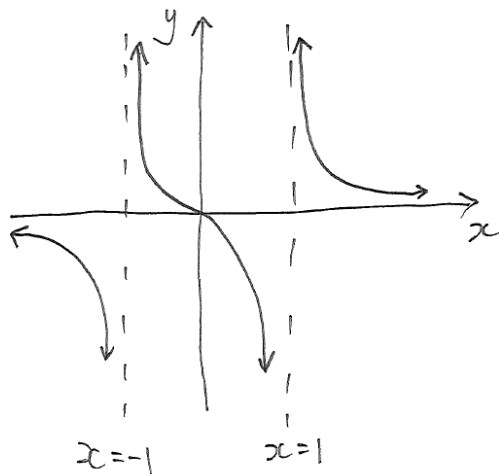
$$= \frac{(x^2-1) \times 2 - 2x \times (2x)}{(x^2-1)^2}$$

$$= \frac{-2-2x^2}{(x^2-1)^2}$$

Now $(x^2-1)^2$ is always positive and the numerator is always < -2 (negative).

Hence the derivative is a negative value for all x .

(iv) Hence the graph:



Q6,

$$(a) \log_{10}(x+4) - \log_{10}(x-5) = 1$$

$$\log_{10} \left(\frac{x+4}{x-5} \right) = 1$$

$$\frac{x+4}{x-5} = 10$$

$$x+4 = 10x-50$$

$$54 = 9x$$

$$x = 6$$

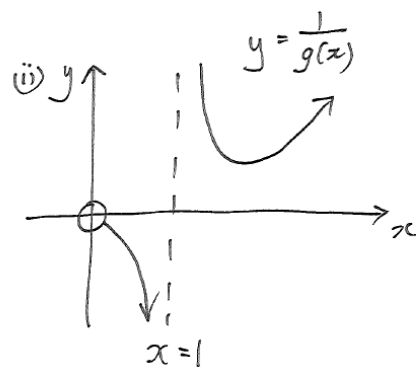
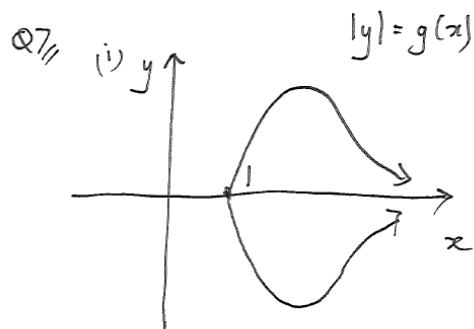
$$(b) 2(4^x) - 17(2^x) + 8 = 0$$

$$\text{Let } m = 2^x \\ m^2 - 17m + 8 = 0$$

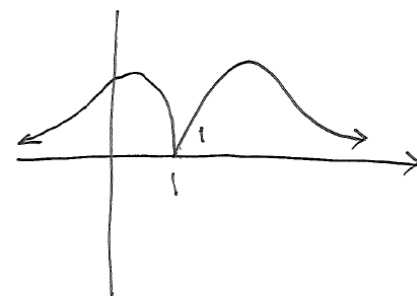
$$2m^2 - 17m + 8 = 0$$

$$(2m-1)(m-8) = 0 \\ m = 8 \text{ or } \frac{1}{2}$$

$$\therefore 2^x = 8 \text{ or } \frac{1}{2} \\ m = 3 \text{ or } -1$$



(iii)



Q8//

$$(a) y = \frac{1}{1+x^2}$$

$$f'(x) = -1(1+x^2)^{-2} \cdot 2x$$

$$= \frac{-2x}{(1+x^2)^2}$$

$$\therefore f'(1) = \frac{-2}{4} = -\frac{1}{2}$$

So m of normal is 2.

$\therefore$ Equation of normal is

$$y - \frac{1}{2} = 2(x-1)$$

$$y = 2x - \frac{1}{2}$$

$$(b) 7x + y = 4$$

$$\text{has } m = -7$$

$$\frac{dy}{dx} = 3x^2 - 12x + 2$$

$$\text{Solve } 3x^2 - 12x + 2 = -7$$

$$3x^2 - 12x + 9 = 0$$

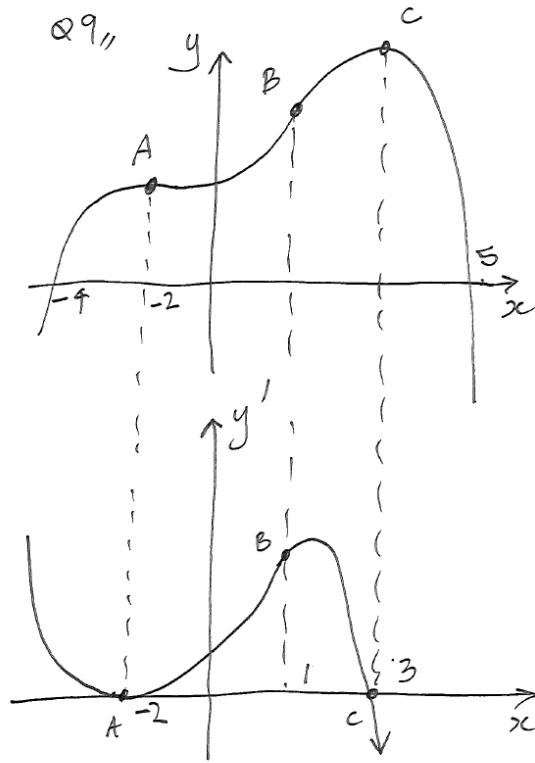
$$x^2 - 4x + 3 = 0$$

$$(x-3)(x-1) = 0$$

$$x = 1 \text{ and } 3$$

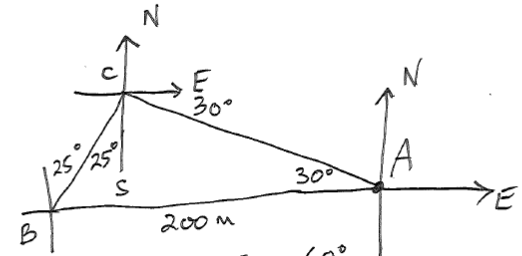
So $(1, -12)$ and $(3, -30)$
are the Required points.

Q9//



Q10//

(i)



$$\text{Now } \angle SCF = 60^\circ$$

$$\text{and } \angle SCB = 25^\circ$$

$$\therefore \angle ACB = 85^\circ, \text{ as required.}$$

(ii) In $\triangle WBC$,

$$\tan 15^\circ = \frac{h}{BC}$$

$$\therefore BC = \frac{h}{\tan 15^\circ}$$

$$\text{and } \tan 75^\circ = \frac{BC}{h}$$

$$\text{So } BC = h \tan 75^\circ, \text{ as required.}$$

(iii) Similarly, $AC = h \tan 67^\circ$

In $\triangle ABC$, by the Cosine Rule,

$$200^2 = (h \tan 75^\circ)^2 + (h \tan 67^\circ)^2 - 2(h \tan 75^\circ)(h \tan 67^\circ) \cos 85^\circ$$

$$40000 = h^2 [\tan^2 75^\circ + \tan^2 67^\circ - 2 \tan 75^\circ \tan 67^\circ \cos 85^\circ]$$

$$40000 = h^2 \times [17.9457]$$

$$h^2 = 2228.946$$

$$h = 47.2 \text{ metres.}$$